

Laboratory Work: SQL DML Queries

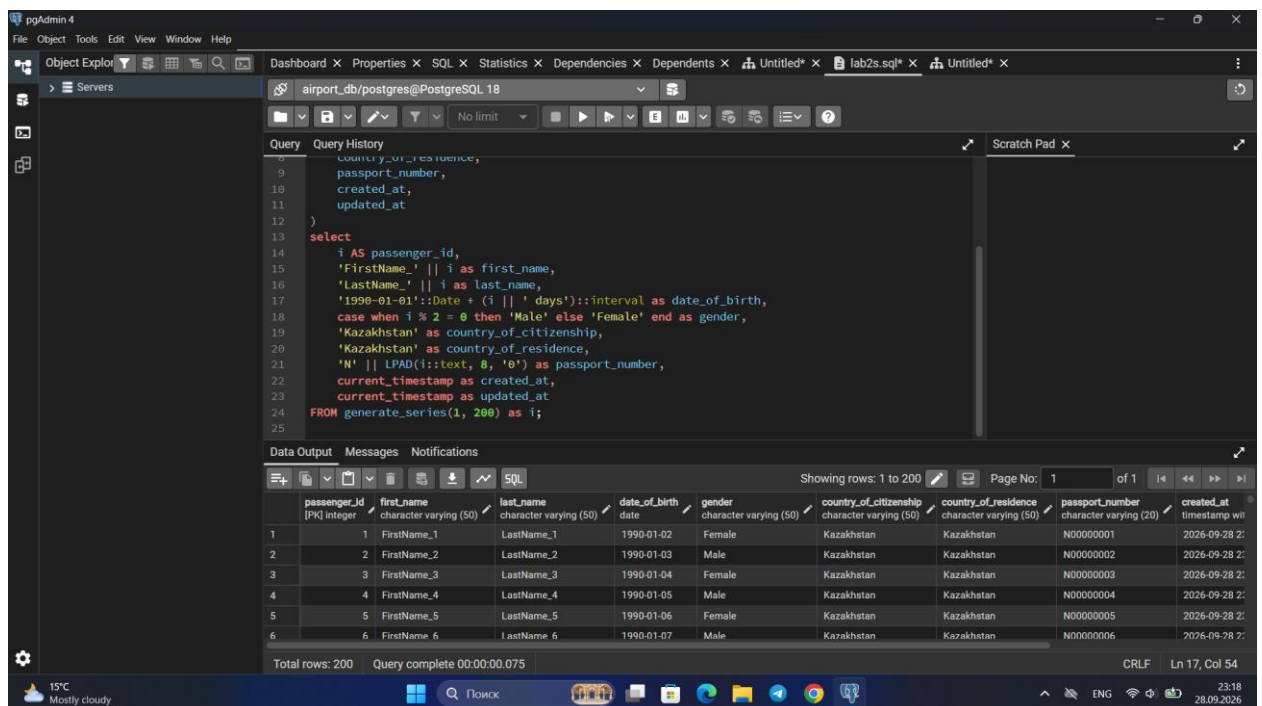
Database Management System — Flight Booking System

Overview & Objectives

This document presents the SQL solutions for Tasks 1 through 15 of the laboratory work. The queries

demonstrate Data Manipulation Language (DML) operations including INSERT, UPDATE, and DELETE actions,

1) In this task, 200 random records were generated and inserted into the airport database. This was done to add test data to the database and check how the tables work with a larger amount of information.



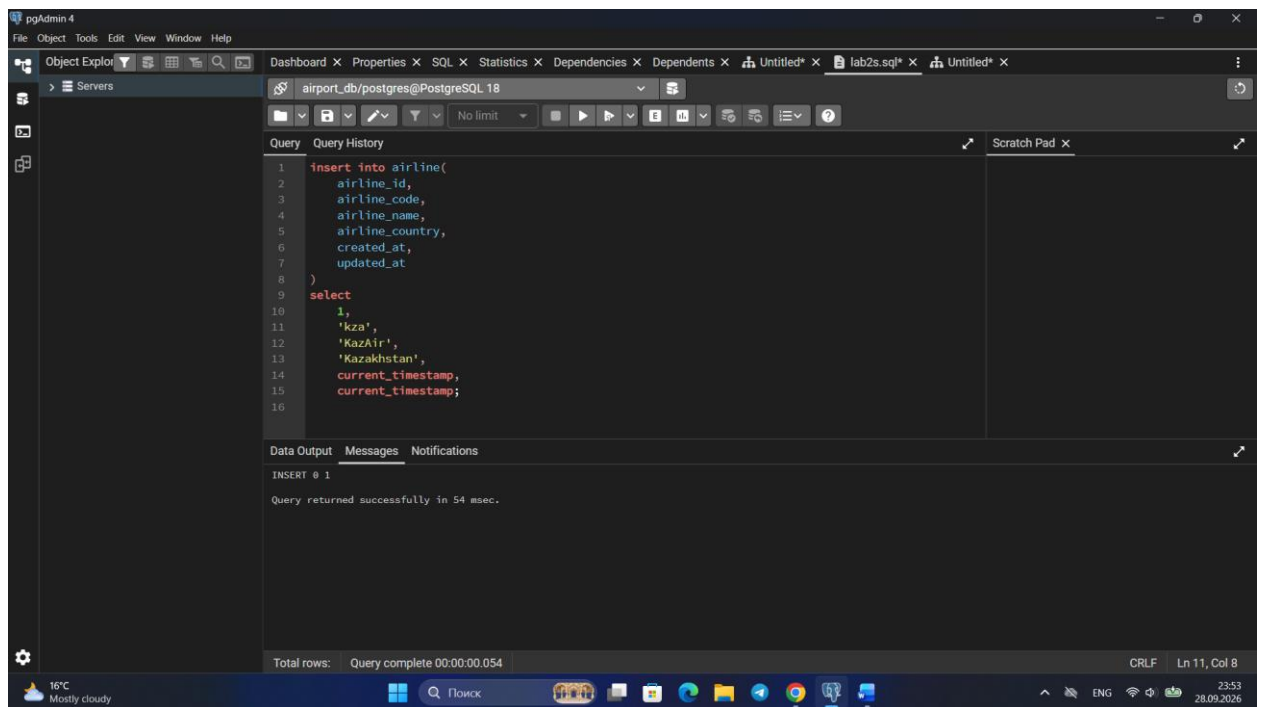
The screenshot shows the pgAdmin 4 interface. The top pane displays a SQL query that generates 200 random records. The bottom pane shows the results of the query, displaying the first 6 rows of the generated data.

```
Query
country_of_residence,
passport_number,
created_at,
updated_at
)
select
1 AS passenger_id,
'FirstName_1' || 1 as first_name,
'LastName_1' || 1 as last_name,
'1990-01-01'::Date + (1 || ' days')::interval as date_of_birth,
case when 1 % 2 = 0 then 'Male' else 'Female' end as gender,
'Kazakhstan' as country_of_citizenship,
'Kazakhstan' as country_of_residence,
'N' || LPAD(1::text, 8, '0') as passport_number,
current_timestamp as created_at,
current_timestamp as updated_at
FROM generate_series(1, 200) as i;
```

passenger_id	first_name	last_name	date_of_birth	gender	country_of_citizenship	country_of_residence	passport_number	created_at
1	FirstName_1	LastName_1	1990-01-02	Female	Kazakhstan	Kazakhstan	N00000001	2026-09-28 21:07:00.000000
2	FirstName_2	LastName_2	1990-01-03	Male	Kazakhstan	Kazakhstan	N00000002	2026-09-28 21:07:00.000000
3	FirstName_3	LastName_3	1990-01-04	Female	Kazakhstan	Kazakhstan	N00000003	2026-09-28 21:07:00.000000
4	FirstName_4	LastName_4	1990-01-05	Male	Kazakhstan	Kazakhstan	N00000004	2026-09-28 21:07:00.000000
5	FirstName_5	LastName_5	1990-01-06	Female	Kazakhstan	Kazakhstan	N00000005	2026-09-28 21:07:00.000000
6	FirstName_6	LastName_6	1990-01-07	Male	Kazakhstan	Kazakhstan	N00000006	2026-09-28 21:07:00.000000

Total rows: 200 Query complete 00:00:00.075

2) A new airline named **KazAir** was added to the airline table. Its initial country was set to **Kazakhstan**.



The screenshot shows the pgAdmin 4 interface. The 'Query' tab is active, displaying the following SQL code:

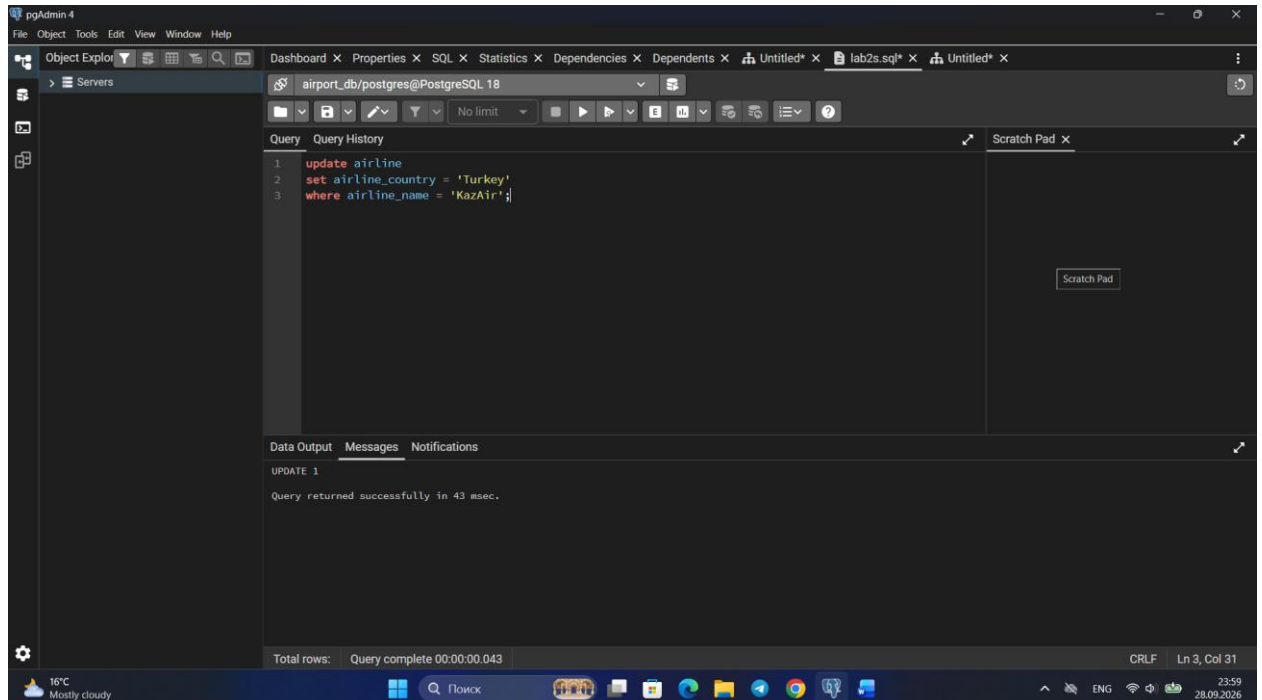
```
1 insert into airline(  
2     airline_id,  
3     airline_code,  
4     airline_name,  
5     airline_country,  
6     created_at,  
7     updated_at  
8 )  
9 select  
10 1,  
11 'kza',  
12 'KazAir',  
13 'Kazakhstan',  
14 current_timestamp,  
15 current_timestamp;  
16
```

The 'Data Output' tab shows the result of the query:

```
INSERT 0 1  
Query returned successfully in 54 msec.
```

The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.054' and 'CRLF Ln 11, Col 8'.

3) The country of the **KazAir** airline was changed from **Kazakhstan** to **Turkey** using the UPDATE command.



The screenshot shows the pgAdmin 4 interface. The 'Query' tab is active, displaying the following SQL code:

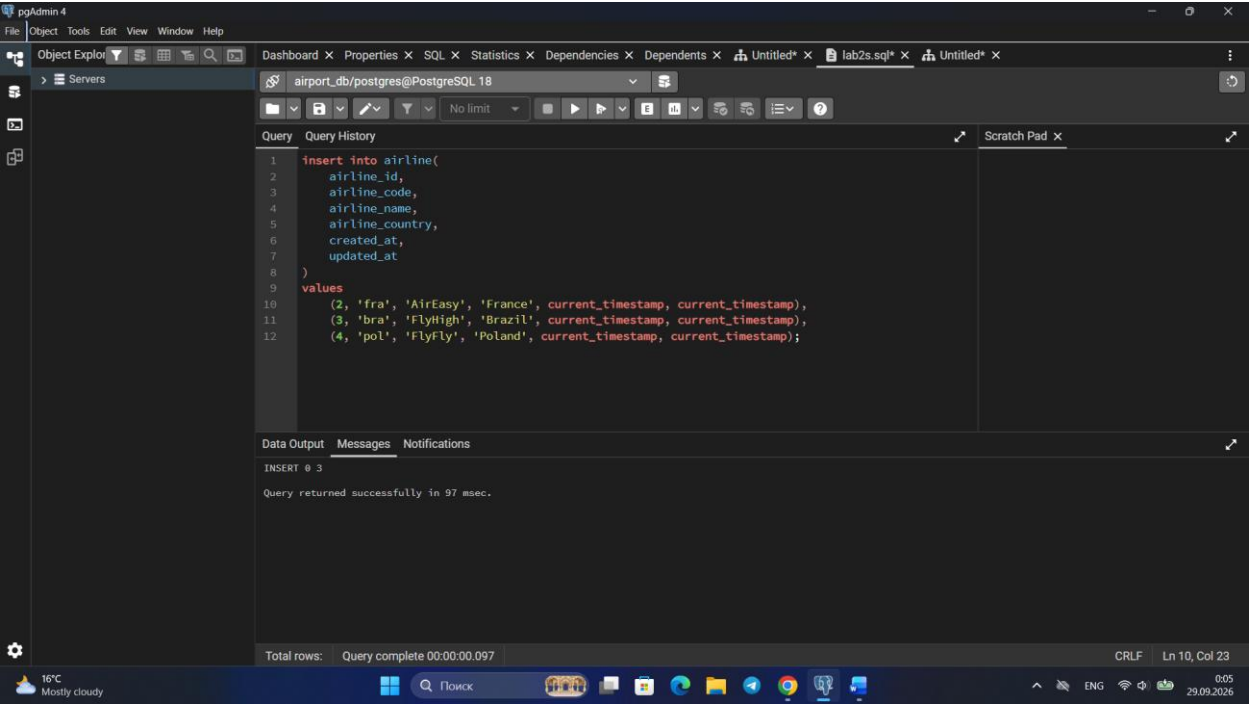
```
1 update airline  
2 set airline_country = 'Turkey'  
3 where airline_name = 'KazAir';
```

The 'Data Output' tab shows the result of the query:

```
UPDATE 1  
Query returned successfully in 43 msec.
```

The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.043' and 'CRLF Ln 3, Col 31'.

4) Three new airlines were inserted into the database at the same time: **AirEasy** from France, **FlyHigh** from Brazil, and **FlyFly** from Poland.



The screenshot shows the pgAdmin 4 interface with the 'airline' table selected in the 'airport_db/postgres@PostgreSQL 18' database. The 'Query' tab is active, displaying an SQL insert statement. The 'Data Output' tab shows the successful execution of the query, returning 3 rows.

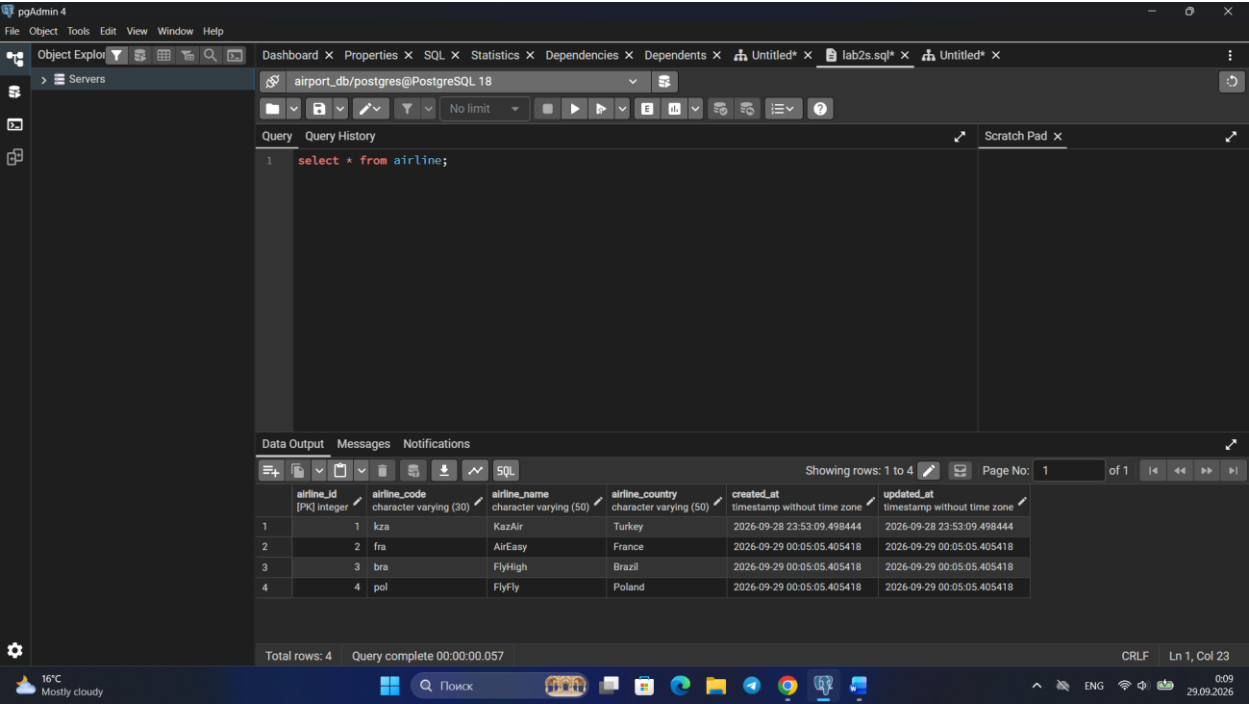
```
1 insert into airline(
2     airline_id,
3     airline_code,
4     airline_name,
5     airline_country,
6     created_at,
7     updated_at
8 )
9 values
10 (2, 'fra', 'AirEasy', 'France', current_timestamp, current_timestamp),
11 (3, 'bra', 'FlyHigh', 'Brazil', current_timestamp, current_timestamp),
12 (4, 'pol', 'FlyFly', 'Poland', current_timestamp, current_timestamp);
```

Data Output Messages Notifications

INSERT @ 3

Query returned successfully in 97 msec.

Total rows: Query complete 00:00:00.097



The screenshot shows the pgAdmin 4 interface with the 'airline' table selected. The 'Query' tab is active, displaying a SELECT query. The 'Data Output' tab shows the result of the query, displaying 4 rows of data.

```
1 select * from airline;
```

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1 of 1

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
1	kza	KazAir	Turkey	2026-09-28 23:53:09.498444	2026-09-28 23:53:09.498444
2	fra	AirEasy	France	2026-09-29 00:05:05.405418	2026-09-29 00:05:05.405418
3	bra	FlyHigh	Brazil	2026-09-29 00:05:05.405418	2026-09-29 00:05:05.405418
4	pol	FlyFly	Poland	2026-09-29 00:05:05.405418	2026-09-29 00:05:05.405418

Total rows: 4 Query complete 00:00:00.057

5) All flights with an arrival date in **2024** were deleted from the database. A condition was used to select only the required records.

pgAdmin 4

Object Explorer | Servers (1) | PostgreSQL 18 | Databases (7) | airport | airport_db | public

Query: `select * from flights;`

Data Output: Showing rows: 1 to 5 | Page No: 1 | of 1

flight_id [PK] integer	sch_departure_time timestamp without time zone	sch_arrival_time timestamp without time zone	departing_airport_id integer	arriving_airport_id integer	departing_gate text	arriving_gate character varying (50)	airline_id integer	act_departure_time timestamp without time zone
1	2024-05-10 10:00:00	2024-05-10 14:00:00	1	1	A1	B1	1	2024-05-10 10:05:00
2	2024-11-20 18:00:00	2024-11-20 22:30:00	1	1	A2	B2	1	2024-11-20 18:00:00
3	2023-06-15 08:00:00	2023-06-15 12:00:00	1	1	C1	D1	1	2023-06-15 08:00:00
4	2025-01-10 15:00:00	2025-01-10 19:00:00	1	1	C2	D2	1	2025-01-10 15:10:00
5	2026-03-01 09:00:00	2026-03-01 13:00:00	1	1	E1	F1	1	2026-03-01 09:00:00

Total rows: 5 | Query complete 00:00:00.076

pgAdmin 4

Object Explorer | Servers (1) | PostgreSQL 18 | Databases (7) | airport | airport_db | public

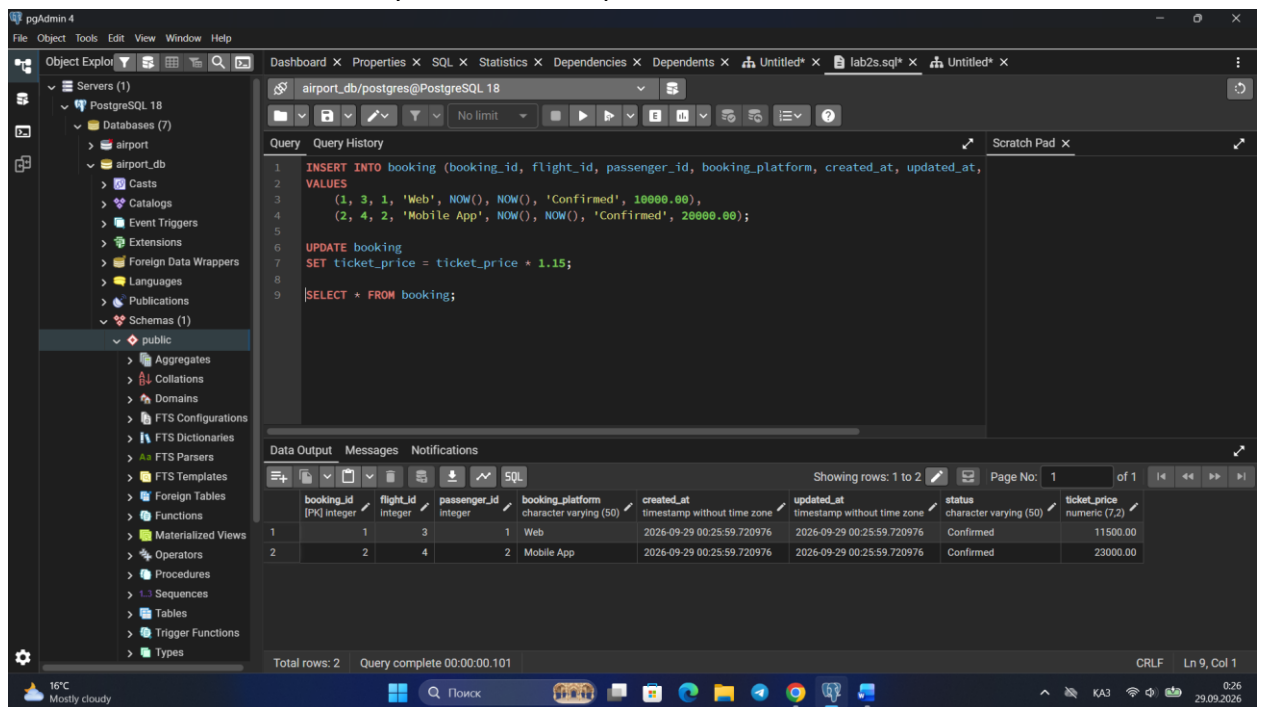
Query: `DELETE FROM flights
WHERE EXTRACT(YEAR FROM sch_arrival_time) = 2024;`

Data Output: Showing rows: 1 to 3 | Page No: 1 | of 1

flight_id [PK] integer	sch_departure_time timestamp without time zone	sch_arrival_time timestamp without time zone	departing_airport_id integer	arriving_airport_id integer	departing_gate text	arriving_gate character varying (50)	airline_id integer	act_departure_time timestamp without time zone
1	2023-06-15 08:00:00	2023-06-15 12:00:00	1	1	C1	D1	1	2023-06-15 08:00:00
2	2025-01-10 15:00:00	2025-01-10 19:00:00	1	1	C2	D2	1	2025-01-10 15:10:00
3	2026-03-01 09:00:00	2026-03-01 13:00:00	1	1	E1	F1	1	2026-03-01 09:00:00

Total rows: 3 | Query complete 00:00:00.083

6) The price of all tickets in the booking table was increased by **15%**. This demonstrates how UPDATE can be used to modify values in multiple records.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure for 'airport_db' under the 'public' schema. The main query editor contains the following SQL code:

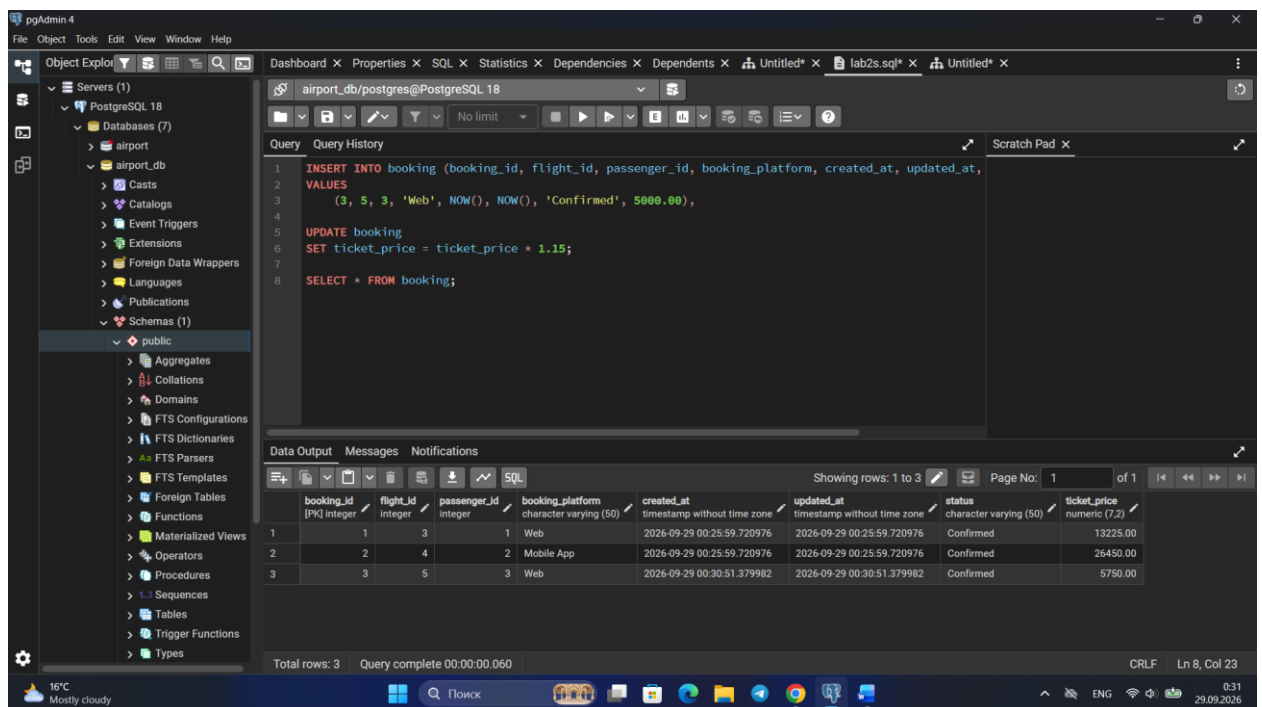
```
1 INSERT INTO booking (booking_id, flight_id, passenger_id, booking_platform, created_at, updated_at,  
2 VALUES  
3 (1, 3, 1, 'Web', NOW(), NOW(), 'Confirmed', 10000.00),  
4 (2, 4, 2, 'Mobile App', NOW(), NOW(), 'Confirmed', 20000.00);  
5  
6 UPDATE booking  
7 SET ticket_price = ticket_price * 1.15;  
8  
9 SELECT * FROM booking;
```

The 'Data Output' tab shows the result of the query, displaying 2 rows of data:

booking_id [PK] integer	flight_id integer	passenger_id integer	booking_platform character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	status character varying (50)	ticket_price numeric (7,2)
1	1	3	1 Web	2026-09-29 00:25:59.720976	2026-09-29 00:25:59.720976	Confirmed	11500.00
2	2	4	2 Mobile App	2026-09-29 00:25:59.720976	2026-09-29 00:25:59.720976	Confirmed	23000.00

Total rows: 2 Query complete 00:00:00.101

7) All tickets with a price of less than **10000** were deleted from the booking table.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure for 'airport_db' under the 'public' schema. The main query editor contains the following SQL code:

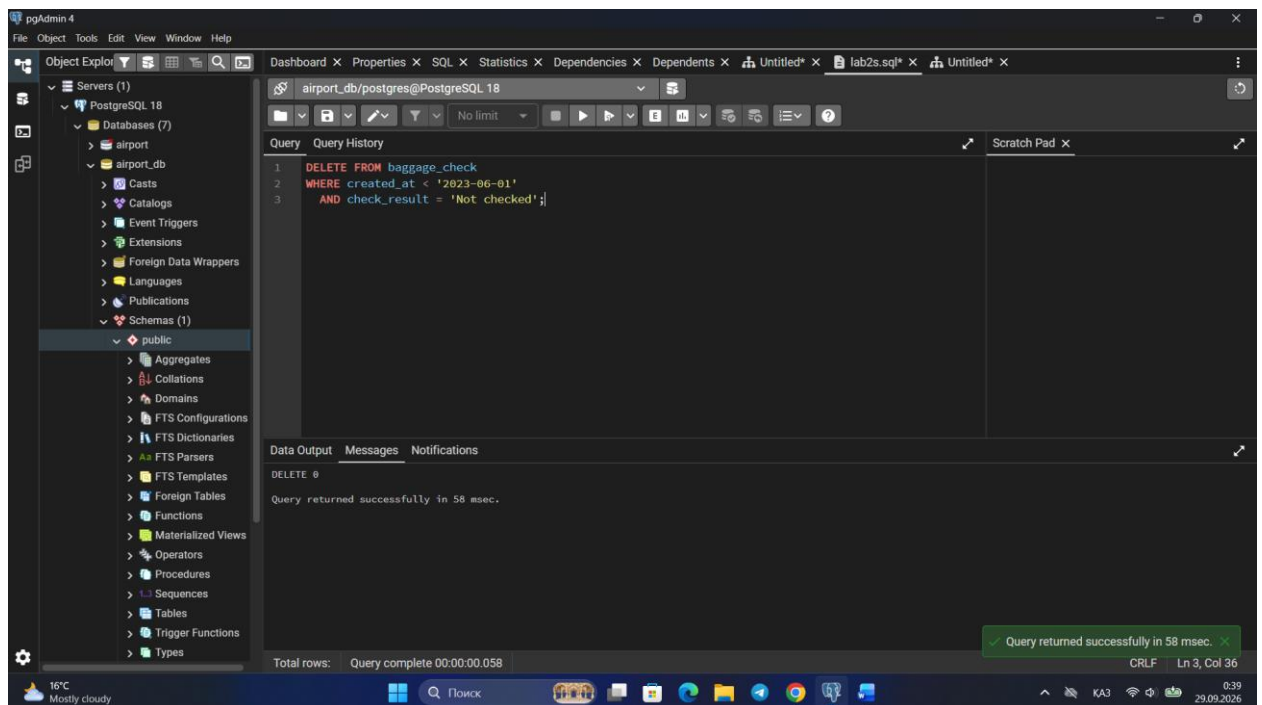
```
1 INSERT INTO booking (booking_id, flight_id, passenger_id, booking_platform, created_at, updated_at,  
2 VALUES  
3 (3, 5, 3, 'Web', NOW(), NOW(), 'Confirmed', 5000.00),  
4  
5 UPDATE booking  
6 SET ticket_price = ticket_price * 1.15;  
7  
8 SELECT * FROM booking;
```

The 'Data Output' tab shows the result of the query, displaying 3 rows of data:

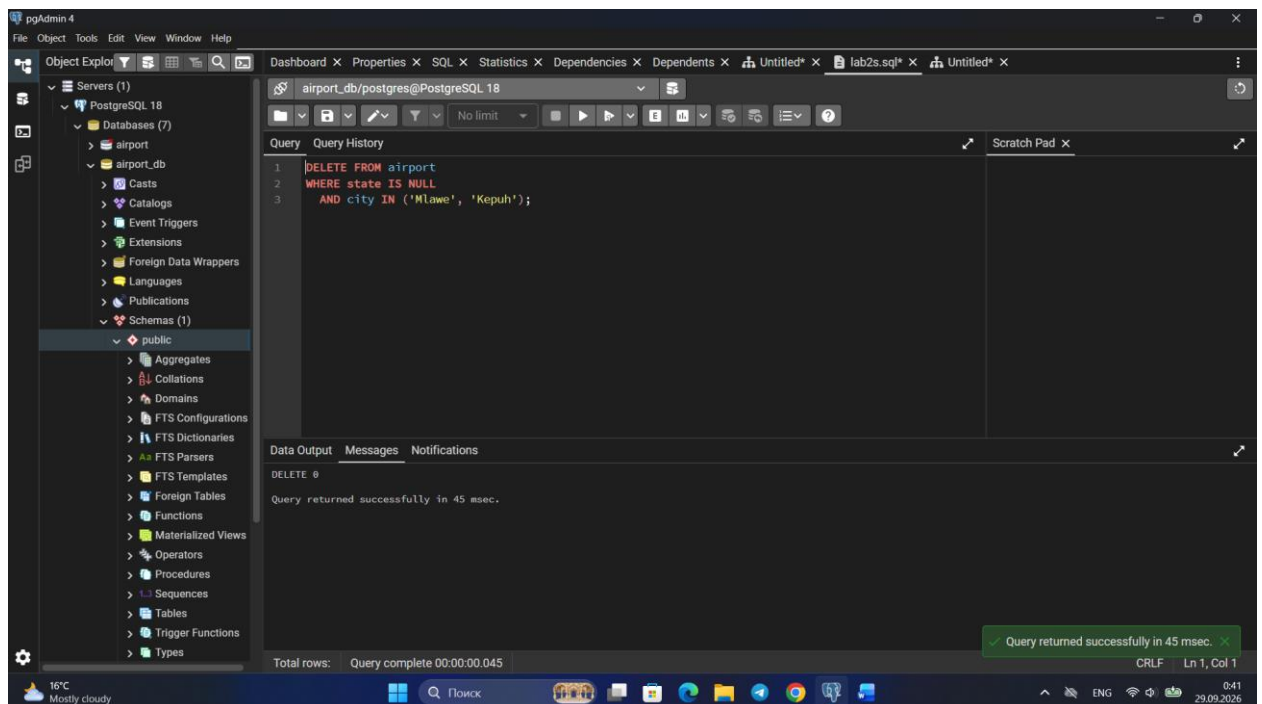
booking_id [PK] integer	flight_id integer	passenger_id integer	booking_platform character varying (50)	created_at timestamp without time zone	updated_at timestamp without time zone	status character varying (50)	ticket_price numeric (7,2)
1	1	3	1 Web	2026-09-29 00:25:59.720976	2026-09-29 00:25:59.720976	Confirmed	13225.00
2	2	4	2 Mobile App	2026-09-29 00:25:59.720976	2026-09-29 00:25:59.720976	Confirmed	26450.00
3	3	5	3 Web	2026-09-29 00:30:51.379982	2026-09-29 00:30:51.379982	Confirmed	5750.00

Total rows: 3 Query complete 00:00:00.060

9) Baggage check records created before **June 1, 2023** with the result '**Not checked**' were deleted from the database.

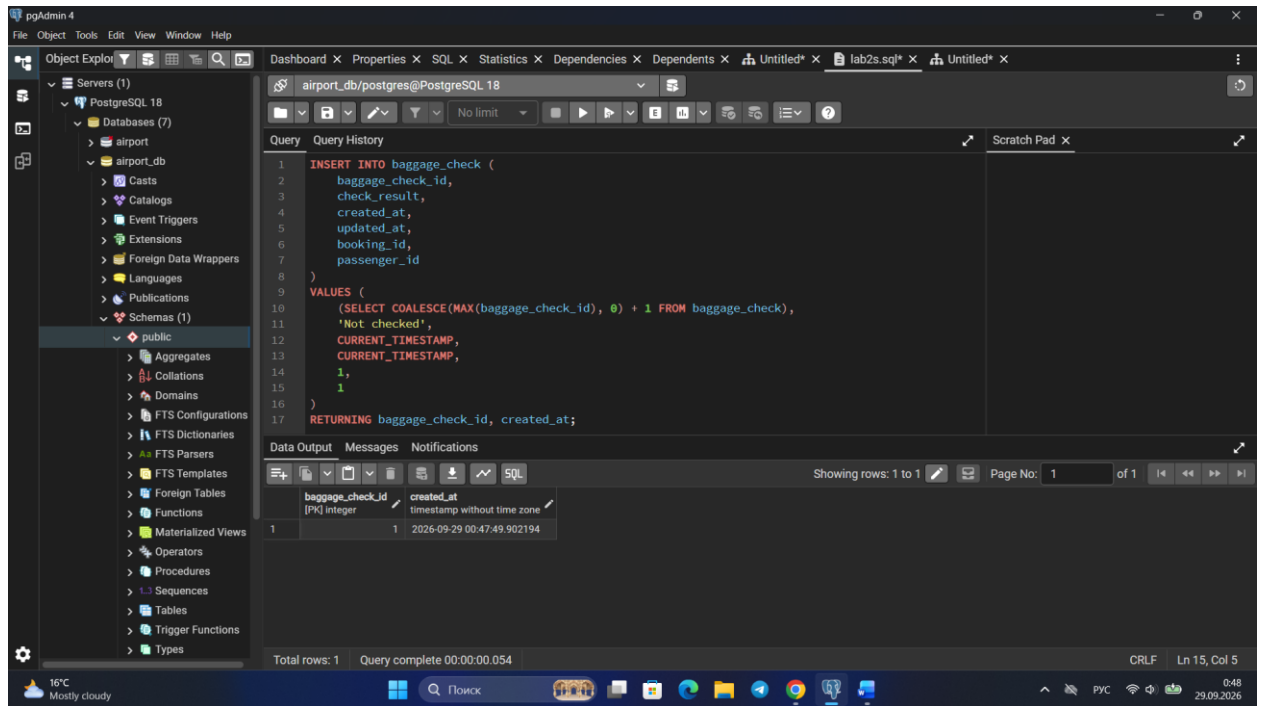


10) Airport records were deleted when the state field was missing and the city was either **Mlawe** or **Kepuh**.



11) A new baggage inspection with the status '**Not checked**' was inserted. The RETURNING command was used to immediately display the generated **baggage_check_id** and **created_at**

values.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure for 'airport_db' in 'PostgreSQL 18'. The main query editor contains the following SQL code:

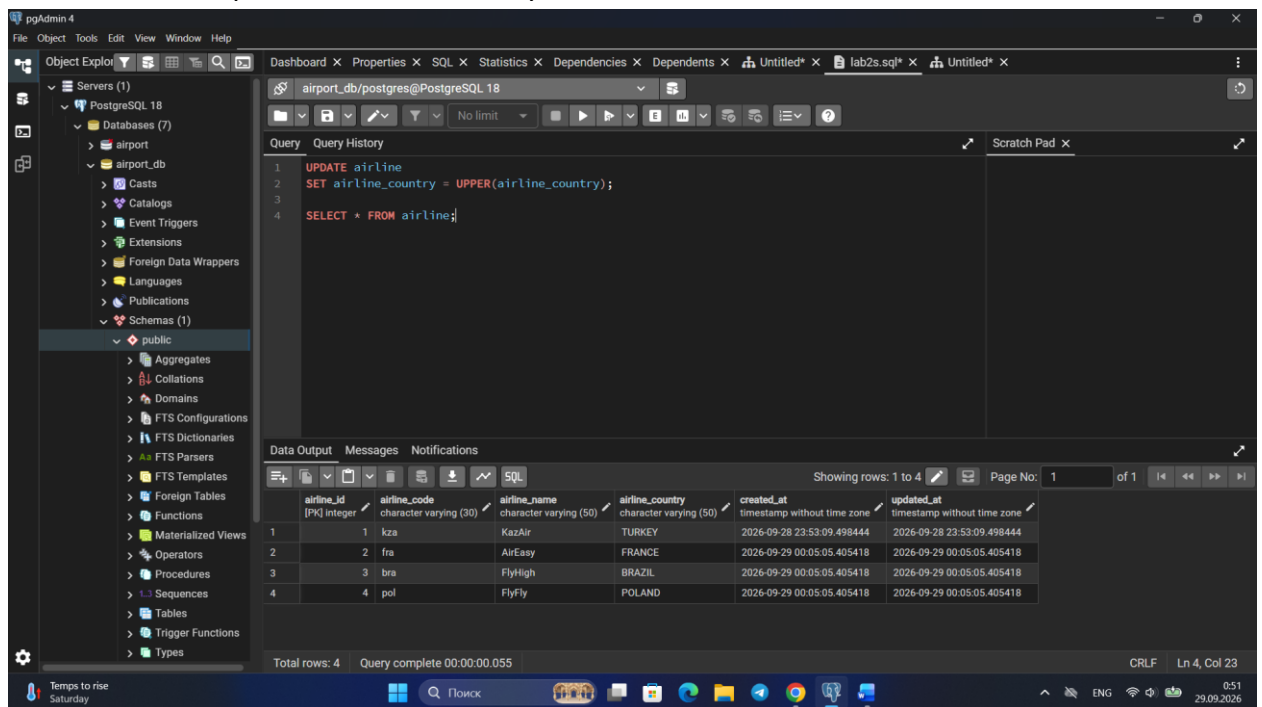
```
1 INSERT INTO baggage_check (
2 baggage_check_id,
3 check_result,
4 created_at,
5 updated_at,
6 booking_id,
7 passenger_id
8 )
9 VALUES (
10 (SELECT COALESCE(MAX(baggage_check_id), 0) + 1 FROM baggage_check),
11 'Not checked',
12 CURRENT_TIMESTAMP,
13 CURRENT_TIMESTAMP,
14 1,
15 1
16 )
17 RETURNING baggage_check_id, created_at;
```

The 'Data Output' tab at the bottom shows the result of the query:

baggage_check_id	created_at
[PK] integer	timestamp without time zone
1	2026-09-29 00:47:49.902194

Total rows: 1 Query complete 00:00:00.054

12) All country names in the airline table were converted to **uppercase** using the UPPER() function. This helps maintain consistency in the database.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure for 'airport_db' in 'PostgreSQL 18'. The main query editor contains the following SQL code:

```
1 UPDATE airline
2 SET airline_country = UPPER(airline_country);
3
4 SELECT * FROM airline;
```

The 'Data Output' tab at the bottom shows the result of the query:

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
[PK] integer	character varying (30)	character varying (50)	character varying (50)	timestamp without time zone	timestamp without time zone
1	kza	KazAir	TURKEY	2026-09-28 23:53:09.498444	2026-09-28 23:53:09.498444
2	fra	AirEasy	FRANCE	2026-09-29 00:05:05.405418	2026-09-29 00:05:05.405418
3	bra	FlyHigh	BRAZIL	2026-09-29 00:05:05.405418	2026-09-29 00:05:05.405418
4	pol	FlyFly	POLAND	2026-09-29 00:05:05.405418	2026-09-29 00:05:05.405418

Total rows: 4 Query complete 00:00:00.055

13) The airline with airline_id = 5 was updated. Its name was changed to **Global Airways**, its country to **United Kingdom**, and the update_at field was set to the current date.

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure for 'airport_db' in 'PostgreSQL 18'. The main pane shows an SQL query in the 'Query' tab:

```
1 UPDATE airline
2 SET airline_name = 'Global Airways',
3     airline_country = 'United Kingdom',
4     updated_at = CURRENT_DATE
5 WHERE airline_id = 5;
6
7 SELECT * FROM airline WHERE airline_id = 5;
```

The 'Data Output' tab shows the result of the query:

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
5	OLD	Global Airways	United Kingdom	2026-09-29 00:52:30.634432	2026-09-29 00:00:00

The status bar indicates 'Total rows: 1' and 'Query complete 00:00:00.079'.

14) The state field was set to 'Capital District' for airports located in **Astana**, **London**, and **Tokyo**.

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure for 'airport_db' in 'PostgreSQL 18'. The main pane shows an SQL query in the 'Query' tab:

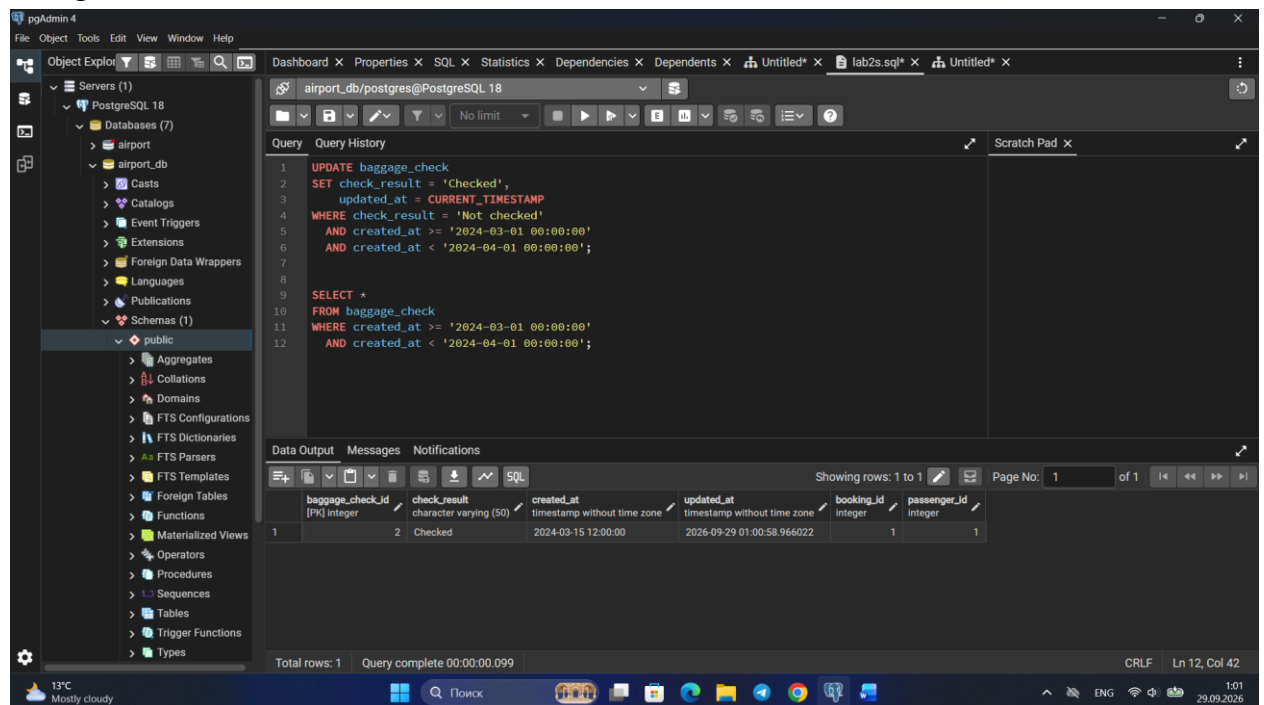
```
1 UPDATE airport
2 SET state = 'Capital District',
3     updated_at = NOW()
4 WHERE city IN ('Astana', 'London', 'Tokyo');
5
6 SELECT * FROM airport WHERE city IN ('Astana', 'London', 'Tokyo');
```

The 'Data Output' tab shows the result of the query:

airport_id	airport_name	country	state	city	created_at	updated_at
1	Astana International Airp...	Kazakhstan	Capital District	Astana	2026-09-29 00:22:42.288556	2026-09-29 00:56:22.557051
2	Astana Airport	Kazakhstan	Capital District	Astana	2026-09-29 00:55:13.162049	2026-09-29 00:56:22.557051
3	London Airport	United Kingdom	Capital District	London	2026-09-29 00:55:13.162049	2026-09-29 00:56:22.557051
4	Tokyo Airport	Japan	Capital District	Tokyo	2026-09-29 00:55:13.162049	2026-09-29 00:56:22.557051

The status bar indicates 'Total rows: 4' and 'Query complete 00:00:00.097'.

15) All baggage check records created in **March 2024** with the status '**Not checked**' were changed to '**Checked**'.



Conclusion: During this laboratory work, I practiced key PostgreSQL DML operations—INSERT, UPDATE, DELETE, and the RETURNING clause—using an airport database. I gained hands-on experience inserting new records, modifying data using functions and conditions, deleting records, and retrieving generated values. Overall, this lab strengthened my practical skills in managing and manipulating data within a relational database system.